

# CS 240: AI/ML Lab

Sunita Sarawagi

# Lab schedule

- Introductory lab on data handling and KNN classifiers.
- Probability/matrix-vector calculations, simple ML pipeline and Kaggle
- Linear regression (closed form), Regularization, lasso/ridge regression
- Kernel regression
- Logistic regression (using gradient descent)
- SVMs
- Decision Trees
- Ensemble classifiers, boosting/bagging, random forests
- PCA, dimensionality reduction, Clustering, k-means
- Feedforward NNs + backprop (vectors)
- CNN-based classifiers (images)
- Simple NLP, text-based classification using embeddings
- RNNs, Transformers, and LLMs (sequence)

# Lab Evaluation

- Grading scheme (Approximate)

- Graded labs: 30% (best n-2 of n) individual
  - Graded in the first half on the topic of the previous week (starts from Week-2)
  - Ungraded second half on next week's topic.
- Lab HWs: 20% (4) to be done in groups of 2, LLM-usage allowed but with disclosure of prompt. Random vivas will be conducted to test ownership and assimilation.
- Mid-term exam: 25%
- Final exam: 25%

- No internet in labs

- How will labs be graded: auto-graded

- Attendance is mandatory. DX grade if attendance less than 75%.

- Labs will be in python. Posted on Moodle.

# Plagiarism policy

- No internet usage during labs, unless explicitly allowed
- Copying from other students during labs is strictly prohibited.
- Violations will be reported to DDAC which incurs severe penalties for both parties.
- HWs: LLM usage allowed but submit prompt. You are expected to understand the submission. Random vivas --  
- strict penalty if not able to explain.

# TAs

|                                 |         |
|---------------------------------|---------|
| Darshan Prabhu (24D0362)        | PhD     |
| Anvay Vipul Shah (22D0368)      | PhD     |
| Suvasish Das (24D0377)          | Masters |
| Mayukh Mondal (25D0363)         | Masters |
| Shubham Dagar (25M2102)         | Masters |
| Abhay (25M2141)                 | Masters |
| Om Ketan Hotkar (24M2127)       | Masters |
| Gaurav Kumar (24M0786)          | Masters |
| Shashank Vinod Zanzad (24M0780) | Masters |
| Aniket Ashwin Waghela (25M2104) | Masters |
| Chinthaparthi Babu Sujan Reddy  | UG3     |
| Niral Charan (23b1045)          | UG3     |
| Yash Santosh Jonjale (22B0990)  | UG4     |
| Aditya Agrawal (22B0955)        | UG4     |
| Abhi Jain (23b0903)             | UG3     |
| Malay Kedia (23B1034)           | UG3     |
| Rishi Kalra (23b1081)           | UG3     |